Recursive LLM Ecosystems Show an Inheritance Shock, Not a Simple Minimum Viable Population Threshold

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Abstract

We study whether a recursive large language model (LLM) ecosystem has a minimum viable population of sufficiently distinct agents below which it collapses. Prior work on model collapse focuses on single recursive training loops, while the population-level question remains open and even the meaning of an "individual" agent is underspecified. We address this gap with a bounded recursive inheritance experiment that uses a frontier API model as the generator, behavioral distance to define individuality, and four generations of synthetic handoff under both replacement and accumulation regimes.

Our setup evaluates six conditions that vary population size, prompt-level role diversity, and access to a persistent human anchor. Across 576 recursive outputs, we find no clean threshold in which larger populations reliably avoid collapse. Instead, the dominant pattern is an immediate generation-1 truthfulness shock across every condition, with truth rates dropping from 0.67–0.92 at generation 0 to 0.42–0.50 at generation 1, followed by partial recovery to 0.67–0.83 by generation 3. Role-diverse agents are behaviorally distinct by our assay, with an individuality ratio of 1.1873 versus 0.9294 for homogeneous agents, but they do not protect the ecosystem. Final-generation collapse severity is positively correlated with the individuality proxy ($\rho = 0.8281$, $p = 0.0418$), and the best final condition is the single-agent accumulation regime.

These results argue against a universal minimum viable population threshold in this proxy setting. They instead point to inheritance quality and preserved human grounding as the more important determinants of recursive stability.

1 Introduction

Recursive synthetic text is no longer a speculative concern. Large language models already write summaries, code, documentation, and conversational data that later systems may absorb. When this happens repeatedly, an information ecosystem can degrade before standard task metrics make the failure obvious. Prior work on model collapse shows that low-probability modes disappear early under recursive self-training, even when headline quality appears stable Shumailov et al. [2023], Briesch et al. [2023]. Our main question is simple: how many sufficiently distinct agents are needed to keep a recursive LLM ecosystem from collapsing?

This question matters because population count is easy to name but hard to interpret. Ten chat sessions of the same model do not necessarily give ten independent producers. On the other hand, small differences in role, memory, or data access may create meaningfully different behaviors. Existing collapse work largely studies a single model loop or compares replacement against accumulation in a single lineage Alemohammad et al. [2023], Gillman et al. [2024], Gerstgrasser et al. [2024]. It does not tell us whether there is a population-level analogue of minimum viable population, nor does it give a practical definition of what counts as a distinct individual in an LLM ecosystem.

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We address that gap with a bounded recursive inheritance experiment. Instead of retraining descendant weights, which was unstable in the available local stack, we let later-generation agents inherit exemplars from earlier synthetic outputs while varying three factors: nominal population size, prompt-level distinctness, and whether a small human anchor persists across generations. We also define individuality behaviorally: an agent counts as distinct when its between-agent output distance exceeds its own within-agent stochastic variance on the same prompts. This lets us separate nominal population size from effective population size.

Figure 1 previews the central result. We do not observe a clean threshold where larger populations reliably prevent collapse. Instead, every condition suffers a strong generation-1 truthfulness shock, then partially recovers over the next two generations. The best final condition is P1-HOMOGENEOUS-ACCUMULATION with a collapse score of -0.0072, while the worst is P4-ROLE-DIVERSE-ACCUMULATION with a collapse score of 0.3147. Prompt-level role diversity makes agents more distinguishable, raising the individuality ratio from 0.9294 to 1.1873, but it does not improve recursive robustness.

Our contributions are:

- We propose a behavioral definition of LLM individuality based on between-agent distance relative to within-agent stochastic variance.
- We conduct a six-condition recursive inheritance study that varies population size, prompt-level distinctness, and replacement versus accumulation across four generations.
- We show that the dominant failure mode in this setting is a generation-1 truthfulness shock with partial recovery, rather than smooth monotonic collapse.
- We show that prompt-level role diversity is measurable but not protective: more behaviorally distinct agents can still yield worse final collapse scores.

The rest of the paper is organized as follows. Section 2 situates our study within recursive training, anti-collapse interventions, and population diversity work. Section 3 describes the recursive inheritance proxy, individuality assay, and evaluation protocol. Section 4 presents the main findings, and Section 5 discusses their interpretation, limitations, and implications.

2 Related Work

Recursive self-training and model collapse. The central observation from the model collapse literature is that recursive training on generated data can erase distributional support before broad task quality visibly fails. Shumailov et al. show that repeated training on generated data drives tail loss and distribution drift across multiple modalities Shumailov et al. [2023]. Alemohammad et al. describe related self-consuming dynamics and emphasize the role of fresh real data in preventing autophagy Alemohammad et al. [2023]. For language specifically, Briesch et al. find that diversity deteriorates long before correctness collapses in a self-consuming loop Briesch et al. [2023]. We build directly on this diversity-first framing, but we ask a different question: whether collapse behavior changes when the recursive loop is treated as a population rather than a single lineage.

Accumulation and correction as anti-collapse mechanisms. Several recent papers argue that collapse is not inevitable once the data regime changes. Gillman et al. show that corrective operators can stabilize recursive generative training Gillman et al. [2024]. Gerstgrasser et al. give a direct counterexample to inevitability by showing that accumulation of real and synthetic data can remain stable when replacement fails Gerstgrasser et al. [2024]. These papers motivate one of our core design choices. We treat replacement and accumulation as first-order experimental factors, because any claim about a minimum viable population is confounded if human refresh is ignored. Our results support this emphasis on data governance, although in our bounded setting accumulation helps only modestly on average.

Population diversity and operational individuality. The minimum viable population idea comes from biology and ecology, but the closest operational precedent for our setting comes from multi-agent learning. McKee et al. show that population and environment diversity can affect held-out generalization, and they measure behavioral diversity rather than assuming it from nominal counts McKee et al. [2021]. That distinction is crucial for LLM ecosystems. Agent-society and role-playing work such as CAMEL, Generative Agents, and RoleLLM provides mechanisms for constructing persistent or role-conditioned differences, but these papers do not study recursive collapse itself Li

et al. [2023], Park et al. [2023], Wang et al. [2023]. We borrow their intuition that roles and memory can define agents behaviorally, then test whether that distinctness is enough to protect a recursive ecosystem.

Positioning our work. Our paper sits between recursive collapse studies and multi-agent diversity studies. Unlike prior single-lineage collapse work, we vary population size and prompt-level distinctness explicitly. Unlike agent-society work, we focus on recursive inheritance stability rather than task coordination. The result is a population-level proxy experiment that offers two concrete outputs: an operational answer to what counts as an individual, and direct evidence that prompt-level distinctness alone does not create a simple minimum viable population threshold.

3 Methodology

Research question and design. We ask whether a recursive LLM ecosystem has a minimum viable population of sufficiently distinct producers below which it collapses, and whether that threshold survives when a small human anchor is retained. The original plan was to fine-tune local descendant models across generations. We replaced that with a recursive inheritance proxy because the local `torch`/`transformers` stack was not stable enough for a reliable one-session run. In our proxy, generation-0 agents are anchored on human exemplars, and later generations inherit exemplars from prior synthetic outputs. This keeps real LLM behavior in the loop while weakening the claim from weight-space retraining to cultural inheritance.

Models, data, and generation protocol. All text generation uses GPT-4.1-MINI with temperature 0.9 and a maximum of 170 output tokens. We measure embedding distances with TEXT-EMBEDDING-3-SMALL. Human prompts come from OASST1, the persistent human anchor comes from WIKITEXT-103 RAW, and held-out evaluation uses TRUTHFULQA GENERATION. Each condition runs for four generations ($g \in \{0, 1, 2, 3\}$). In every generation, the system produces 12 corpus samples from OASST1 prompts and 12 TRUTHFULQA answers, for 24 outputs per condition-generation cell and 576 recursive outputs overall.

Experimental conditions. We evaluate six conditions:

- P1-HOMOGENEOUS-REPLACEMENT
- P2-HOMOGENEOUS-REPLACEMENT
- P4-HOMOGENEOUS-REPLACEMENT
- P4-ROLE-DIVERSE-REPLACEMENT
- P1-HOMOGENEOUS-ACCUMULATION
- P4-ROLE-DIVERSE-ACCUMULATION

Replacement means later generations inherit only synthetic exemplars. Accumulation means they retain a small human anchor alongside synthetic inheritance. The homogeneous conditions use same-role agents that differ only through sampling noise, while the role-diverse conditions vary prompt role and framing.

What counts as an individual? Our population definition is behavioral rather than architectural. Let d_{within} be the mean distance between repeated outputs from the same agent on the same prompt set, and let d_{between} be the mean distance between outputs from different agents on that set. We define the individuality ratio as

$$\text{IR} = \frac{d_{\text{between}}}{d_{\text{within}}}.$$

An agent population counts as behaviorally distinct when $\text{IR} > 1$, meaning between-agent differences exceed within-agent stochastic variance. This criterion follows the logic that separate API calls should not count as separate individuals unless they are distinguishable beyond sampling noise.

Evaluation metrics. We measure collapse with three families of metrics. First, we track lexical and distributional diversity using distinct-2, distinct-3, token entropy, self-BLEU, and mean embedding similarity. Second, we estimate truthfulness by comparing answer embeddings against correct and incorrect TRUTHFULQA references and converting those comparisons into a binary truthfulness proxy. Third, we compute a composite collapse score that combines relative changes in diversity, truthfulness, and embedding concentration with respect to generation 0. Lower collapse scores are better.

Table 1: Behavioral individuality assay for the $P{=}4$ homogeneous and role-diverse populations. An individuality ratio above 1 means between-agent differences exceed within-agent stochastic variance.

Condition	Within distance	Between distance	Individuality ratio
Homogeneous $P{=}4$ assay	0.0672	0.0625	0.9294
Role-diverse $P{=}4$ assay	**0.0613**	**0.0728**	**1.1873**

Condition	Final collapse score ↓	Final truth rate ↑	Final distinct-2 ↑	Final token entropy ↑
P1-HOMOGENEOUS-ACCUMULATION	**-0.0072**	0.7500	0.9095	8.7962
P4-HOMOGENEOUS-REPLACEMENT	0.0190	**0.8333**	0.9041	8.8324
P1-HOMOGENEOUS-REPLACEMENT	0.2003	0.6667	**0.9394**	**8.9622**
P2-HOMOGENEOUS-REPLACEMENT	0.2056	0.7500	0.9352	8.8352
P4-ROLE-DIVERSE-REPLACEMENT	0.2238	0.6667	0.9297	8.8834
P4-ROLE-DIVERSE-ACCUMULATION	0.3147	0.6667	0.9193	8.8540

Table 2: Final-generation comparison across all recursive ecosystem conditions. Lower collapse score is better. Best values in each column are shown in **bold**. The best overall condition is the single-agent accumulation regime, while the worst is the role-diverse accumulation regime.

Baselines and analysis. The most pessimistic tested baseline is P1-HOMOGENEOUS-REPLACEMENT, which combines the smallest nominal population with full synthetic replacement. The strongest human-anchor baseline is P1-HOMOGENEOUS-ACCUMULATION. The role-diverse $P{=}4$ conditions test whether behavioral distinctness changes the recursive trajectory. For statistical analysis, we report the final-generation Spearman correlation between the individuality proxy and collapse severity across conditions. Because this experiment is deliberately bounded to 12 corpus prompts and 12 TRUTHFULQA questions per cell, we treat the results as exploratory and focus on effect direction, rank ordering, and trajectory shape rather than precise interval estimates.

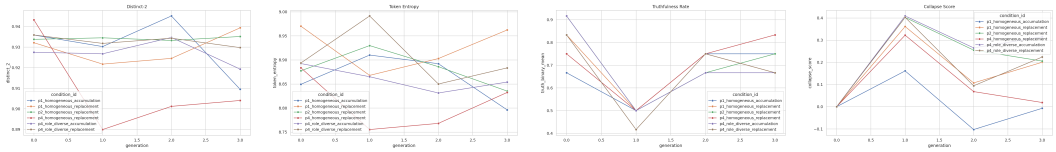
Implementation details. The experiment runs in a Python 3.12.8 virtual environment using the OpenAI Python SDK 2.32.0. The workspace includes four NVIDIA RTX A6000 GPUs, but the final API-based inheritance pipeline does not use them. One end-to-end caveat is that the first full generation run predates a determinism patch that sorted inherited exemplars. The saved raw outputs are stable and the analysis reruns identically, but a bitwise-identical re-execution should use the patched script.

4 Results

Behavioral individuality is measurable. Table 1 shows that homogeneous prompt variants do not qualify as distinct individuals by our behavioral criterion. Their individuality ratio is 0.9294, because between-agent distance (0.0625) is smaller than within-agent variance (0.0672). Role-diverse variants cross the threshold with an individuality ratio of 1.1873. This gives a concrete operational answer to the population question: separate sessions count as separate individuals only when their outputs diverge more than sampling noise alone would predict.

There is no clean minimum viable population threshold. Table 2 compares the final generation across all six conditions. If a simple threshold existed, we would expect larger populations to dominate smaller ones. We do not observe that pattern. The best final condition is P1-HOMOGENEOUS-ACCUMULATION with a collapse score of -0.0072, while the worst is P4-ROLE-DIVERSE-ACCUMULATION with a collapse score of 0.3147. Grouping by nominal population size yields mean final collapse scores of 0.0966 for $P{=}1$, 0.2056 for $P{=}2$, and 0.1858 for $P{=}4$, which again does not support a monotonic threshold.

The dominant pattern is an inheritance shock at generation 1. Figure 1 shows the clearest temporal regularity in the experiment. Truthfulness starts relatively high at generation 0, ranging from 0.67 to 0.92 across conditions, then drops sharply to 0.42–0.50 at generation 1 in every condition. After that first synthetic handoff, conditions partially recover to 0.67–0.83 by generation 3. This pattern is stronger and more consistent than any ordering by population size. In other words, the first recursive inheritance step matters more than the nominal number of producers.



(a) **Distinct-2.** Lexical diversity changes are condition-specific and do not collapse monotonically.

(b) **Token entropy.** Final entropy remains relatively close across conditions.

(c) **Truthfulness.** Every condition suffers a sharp generation-1 drop followed by partial recovery.

(d) **Collapse score.** Final rank ordering does not follow nominal population size.

Figure 1: Trajectory summary across the four tracked generations. The shared qualitative pattern is an early inheritance shock, especially in truthfulness, rather than universal monotonic degradation across all metrics.

Prompt-level role diversity is detectable but not protective. The role-diverse conditions have a higher final mean collapse score than the homogeneous conditions: 0.2693 versus 0.1044. They also perform worse than the overall replacement mean in one case and worse than the best homogeneous accumulation condition in both cases. The final-generation Spearman correlation between the individuality proxy and collapse severity is $\rho = 0.8281$ with $p = 0.0418$. Because higher collapse scores are worse, this association runs opposite the initial hypothesis that more distinct agents would protect the ecosystem.

Accumulation helps modestly on average, but not universally. When grouped by data regime, accumulation yields a slightly better mean final collapse score than replacement, 0.1538 versus 0.1622. This direction is consistent with prior accumulation-based anti-collapse findings Gerstgrasser et al. [2024]. However, the effect is not universal in our bounded setup. The best condition uses accumulation, but the worst condition does too. This again suggests that preserving a human anchor matters, but its benefit depends on what kind of diversity and inheritance the system preserves.

Embedding concentration still rises over time. Despite partial truthfulness recovery, every condition ends with higher embedding similarity than it started with, which indicates concentration of outputs over time. For example, P1-HOMOGENEOUS-ACCUMULATION rises from 0.1579 to 0.2018, P4-HOMOGENEOUS-REPLACEMENT rises from 0.1580 to 0.2154, and P4-ROLE-DIVERSE-REPLACEMENT rises from 0.1681 to 0.2179. This matches the broader collapse literature in which diversity can degrade even when top-line utility has not yet failed completely Shumailov et al. [2023], Briesch et al. [2023].

5 Discussion

What did we learn about minimum viable population? The main negative result is also the clearest one: this experiment does not support a simple rule such as "below four agents the ecosystem collapses." Population size alone is a weak predictor in our setup. A single-agent accumulation condition finishes best, while several larger populations finish worse. That does not mean population never matters. It means nominal count is not enough without a stronger account of what information each producer contributes and how that information is inherited.

Why did role diversity fail to help? Our individuality assay shows that prompt-level role diversity creates measurable behavioral separation. The issue is that this separation may be mostly stylistic. If distinct agents vary tone, framing, or discourse habits but draw from the same underlying knowledge and the same recursive inheritance channel, they can still transmit the same distortions. The positive correlation between individuality ratio and collapse severity is therefore plausible: more visible stylistic variation can coexist with weaker preservation of factual grounding. In this sense, our results distinguish behavioral distinctness from epistemic diversity.

What does the generation-1 shock imply? The strongest regularity in the paper is the immediate drop after the first synthetic handoff. This suggests that the first transition away from direct human anchoring is a critical point. Later generations do not show universal runaway collapse; instead, they partially recover. A practical implication is that recursive systems may fail first through an

inheritance shock rather than a smooth long-run drift. Monitoring only late-stage degradation would miss that transition.

Limitations. This paper studies a recursive inheritance proxy, not full descendant-model retraining. The experiment is also deliberately small: each condition-generation cell uses 12 corpus prompts and 12 TRUTHFULQA questions, and truthfulness is measured through embedding-based similarity rather than human annotation. In addition, the role-diverse conditions change prompt framing but do not create independent knowledge sources, tools, or model families. Finally, the first full generation run predates a determinism patch that sorts inherited exemplars. These limits mean the paper should be read as evidence about recursive cultural inheritance under bounded conditions, not as a definitive statement about future pretraining pipelines.

Broader implications. The broader message is governance-oriented. If synthetic text is entering shared corpora, then counting the number of contributing agents is an incomplete safety check. What matters more is whether the inheritance mechanism preserves human anchors, external grounding, or genuinely different evidence sources. Systems that optimize for visible surface diversity alone may look healthy while still becoming more concentrated over time.

6 Conclusion

We studied whether recursive LLM ecosystems exhibit a minimum viable population threshold and how to define an individual agent operationally. Using a four-generation recursive inheritance proxy with six conditions, we found no clean threshold in which larger populations reliably prevented collapse. Instead, every condition showed a strong generation-1 truthfulness shock, followed by partial recovery. Our individuality assay gave a concrete answer to the population-definition question: agents count as distinct when between-agent distance exceeds within-agent stochastic variance. By that criterion, role-diverse prompts create distinct individuals, but that distinctness did not protect the ecosystem.

The main takeaway is straightforward. In this bounded setting, recursive stability depends more on inheritance quality and preserved human grounding than on nominal population count alone. Prompt-level role diversity is measurable, but it is not the form of diversity that prevents collapse.

Future work should replace the inheritance proxy with actual across-generation fine-tuning, introduce grounded diversity through retrieval or human refresh, expand the population design to include model-family diversity, and validate failure cases with stronger truthfulness judgments.

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